

APPLIED ANALYTICS: FROM DATA TO DECISIONS | Spring 2026

Final Exam

Onboarding Email
A/B Test Project

Deadline: May 04, 2026 at 11:59 PM

Submission: Upload PDF to Courseworks (Final Exam)

Individual exam

FlowDesk

B2B SaaS
Project Management

FlowDesk Company

FlowDesk is a B2B SaaS company that sells project management software to small and medium businesses (SMBs).

Accounts

A company that signs up for a FlowDesk free trial is called an 'account.' Each account can have multiple users, but for our analysis the account is the key unit.

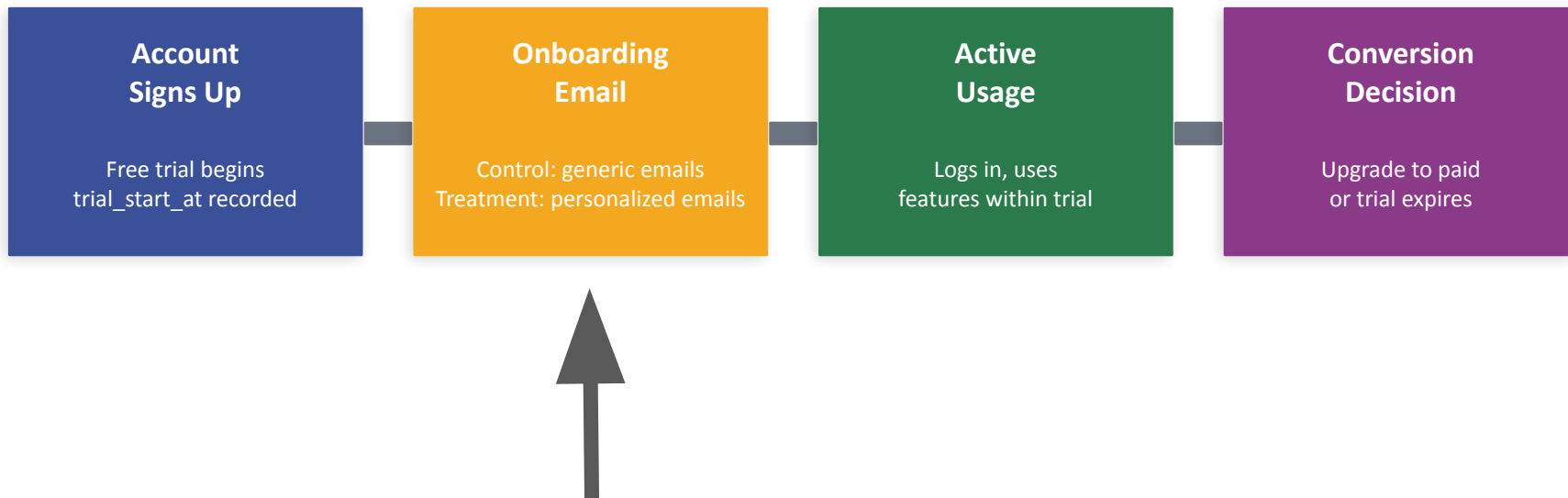
Trial → Paid

Accounts start with a 14-day free trial. The goal of FlowDesk's Growth team is to convert as many accounts as possible into paying subscribers.

You Are...

A Data Analyst on the Growth team. You collaborate with the Marketing team to design, run, and analyze A/B tests on the onboarding experience.

The Account Journey



This is our experiment

The 4 Tables You Have Access To

`fabriziopublic.flowdesk.
dim_account`

One row per account that started a free trial.

`account_id · industry · account_size · trial_start_at · converted_at ·
plan_type · churned_at`

`fabriziopublic.flowdesk.
fact_subscription`

One row per subscription lifecycle event.

`account_id · event_type (trial_start / converted / churned) · event_at ·
mrr (revenue)`

`fabriziopublic.flowdesk.
fact_activity`

One row per account per active day within the trial window.

`account_id · active_date · features_used · n_features`

`fabriziopublic.flowdesk.
experiment_groups`

One row per unit in any experiment.

`experiment_name · unit_id · grouped_at · group`

dim_account — One Row Per Account

<code>account_id</code>	— Unique identifier for the account (unit of analysis)
<code>industry</code>	— Industry of the account (technology, finance, marketing, ...)
<code>account_size</code>	— Employee count bucket: 1-10, 11-50, 51-200, 201-500
<code>trial_start_at</code>	— When the account's free trial began
<code>converted_at</code>	— When the account converted to paid. NULL if never converted.
<code>plan_type</code>	— Subscription plan: starter (\$49/mo), professional (\$99/mo), business (\$199/mo). NULL if not converted.
<code>churned_at</code>	— If a converted account cancelled within the 2-week window. NULL otherwise.

experiment_groups — One Row Per Experiment Unit

<code>experiment_name</code>	— Unique name of the experiment. Always filter to 'onboarding_email_v1'
<code>unit_id</code>	— Account identifier. Joins to dim_account.account_id
<code>grouped_at</code>	— Timestamp when the account was assigned to a group
<code>group</code>	— Group assignment: 'Control' or 'Treatment'

fact_subscription — One Row per subscription lifecycle event.

<code>account_id</code>	— Unique identifier for the account (unit of analysis)
<code>event_type</code>	— "trial_start", "converted", or "churned"
<code>event_at</code>	— Timestamp of the event
<code>mrr</code>	— Monthly Recurring Revenue at event time (for example 'converted'). 0 for trial_start and churned.

fact_activity — One Row per account per active day within the trial window

<code>account_id</code>	— Unique identifier for the account (unit of analysis)
<code>active_date</code>	— Date (YYYY-MM-DD) on which the account had at least one login
<code>features_used</code>	— Comma-separated list of product features used that day
<code>n_features</code>	— Count of distinct features used that day (integer)

EXPERIMENT

The Onboarding Email A/B Test

All the accounts in the experiment are new accounts starting a free trial.

Current State (Control)

All new accounts receive a **generic welcome email sequence** — the same 3 emails sent to every account regardless of what they do during their first session.

New Approach (Treatment)

The Marketing team has developed a **personalized onboarding email sequence** that tailors content to the specific features an account used in its first session. The hypothesis: relevant tips → better engagement → more conversions.

EXPERIMENT

onboarding_email_v1

DATES

Sep 15 – Sep 28, 2025

SPLIT

50% Control / 50% Treatment

DURATION

2 weeks

Your Project

You will produce a single PDF document describing the full experiment, written from three points in time:

Sep 14, 2025

Experiment Planning

- Define unit of randomization and justify the choice
- Define Control and Treatment groups
- Choose and justify 3-5 metrics - at least one must be a ratio metric
- Conduct a power analysis on the pre-experiment period (Sep 1–14, 2025)
- Describe how data will be logged

Sep 15, 2025

Rollout

- Assume the rollout went smoothly

Sep 29, 2025

Analysis &
Recommendation

- Analyze all metrics using data from Sep 15–28 (experiment_name = 'onboarding_email_v1')
- Provide a clear recommendation with justification

Remarks and Tips

① Unit of Randomization

The experiment is randomized at the account level. New accounts (trial_start_at in Sep 15–28) enter the experiment when their trial begins. Accounts behave independently - there is no contamination concern.

② Ratio Metric Requirement

At least one of your metrics must be a ratio metric. Use the delta method to compute its standard error.

③ Power Analysis

Base power analysis on the primary metric using data from Sep 1–14 (accounts whose trial started in that window). Assume a minimum detectable effect of 22% of the baseline. If your baseline is 10 then assume a delta of 22% of 10 = 2.2.

④ fact_activity vs. fact_subscription

fact_activity logs days an account was active during their trial. fact_subscription logs billing lifecycle events. An account can appear in fact_activity without ever converting.

⑤ Experiment Window

For analysis of results, compute all metrics within the 2-week experiment window (Sep 15–28). Any account action (conversion, activity, churn) outside this window should be excluded from your experiment metrics.

Deliverable & Grading

What to Submit

- A single PDF document uploaded to Courseworks
 - Main body: 8 pages or less (US Letter, 11pt or larger)
 - Appendix (no page limit): SQL queries and p-value calculations. You may link p-value calculations to an external spreadsheet
- This is an individual exam - no collaboration allowed
- Your document should be readable by both Marketing managers (high-level results and recommendations) and fellow Data Analysts (technical details, SQL, statistical methods)
- If helpful, you can use the same template structure as in the course example
- **Note:** *The exam is intended to be completed in approximately ~4 hours (but it is not timed, so you can spend more time on it). If you have technical questions about the Final Exam, please post them as a “Reply” in the corresponding Discussion on Courseworks.*

Grading

- Solutions will be posted on May 5th
- The grades will depend on 2 main factors: Technical results and Quality of the presentation (35 points total)
- Grades will be posted by May 14th. If you have questions about your grade, email me within 24 hours from the time grades are posted. 24-hours after grades are posted, course grades will be determined and finalized.

Good luck!

Questions? Post in the Courseworks Discussion
for the Final Exam.

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Applied Analytics
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